Chapter 1

Preliminaries, Background, and Evaluation Metrics

Declaration of Contributions: This chapter was primarily authored by *Nitin Agiwal*, who reviewed and synthesised the mathematical foundations and machine-learning concepts. *Pradyuman Singh Shekhawat* contributed the section on Variational Autoencoders and disentangled representation learning. *Vishesh Gupta* compiled and formalised all fairness evaluation metrics. All team members cross-checked equations and figures for accuracy.

This chapter lays the conceptual and mathematical groundwork required to understand the fairness-aware machine-learning techniques surveyed in subsequent chapters. We begin with the essential mathematical building blocks—drawing on analysis and probability—then introduce the AI/ML architectures that appear throughout the literature, and finally define the fairness metrics used to quantify bias.

1.1 Mathematical Foundations

1.1.1 Sets, Functions, and Notation

Throughout this report we adopt the following notation. Scalars are denoted by lower-case italic letters (x, y, z); vectors by bold lower-case $(\mathbf{x}, \mathbf{z})$; and matrices by bold upper-case $(\mathbf{W}, \mathbf{X})$. The set of real numbers is $\mathbb{R}$, and $\mathbb{R}^d$ denotes a d-dimensional real vector space. We write $\mathcal{X}$ for an input space and $\mathcal{Z}$ for a latent (hidden) space.

A *random variable* X defined over a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ takes values in $\mathcal{X}$. Its *probability distribution* is denoted P_X or simply P. The *expectation* of a function $f(X)$ is

$$\mathbb{E}_{X\sim P}\big[f(X)\big] \;=\; \int_{\mathcal{X}} f(x)\,\mathrm{d}P(x).$$

1.1.2 Infimum and Supremum

Two operations appear repeatedly in the loss functions of the papers we review.

- The **infimum** (inf) of a set $S \subseteq \mathbb{R}$ is its *greatest lower bound*: the largest value ℓ such that $s \geq \ell$ for every $s \in S$. For a function $f : \Lambda \to \mathbb{R}$ we write $\inf_{\lambda \in \Lambda} f(\lambda)$ to mean the smallest value f can approach (not necessarily attained).
- The **supremum** (sup) is the *least upper bound*, analogously defined.

These concepts are essential when the minimum or maximum of a function is not attained—a situation that arises naturally in risk-based fairness objectives (see Section 1.1.7).

1.1.3 Kullback-Leibler Divergence

Given two probability distributions P and Q over the same space, the *Kullback–Leibler (KL) divergence* measures how much P diverges from Q:

$$D_{\mathrm{KL}}(P\|Q) \;=\; \mathbb{E}_{x\sim P}\left[\log \frac{P(x)}{Q(x)}\right] \;=\; \int P(x)\,\log \frac{P(x)}{Q(x)}\,\mathrm{d}x. \tag{1.1}$$

$D_{\mathrm{KL}}(P\|Q) \geq 0$ always, with equality iff $P = Q$ almost everywhere. KL divergence is *asymmetric*: in general $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$.

Role in VAEs. In Variational Autoencoders (Section 1.5), the KL divergence acts as a regulariser that pushes the approximate posterior $q_\Phi(\mathbf{z}|\mathbf{x})$ towards the standard Gaussian prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$.

1.1.4 Total Correlation

Total Correlation (TC), introduced by Watanabe [1], generalises mutual information to measure the overall statistical dependency among n random variables $z_1, z_2, \ldots, z_n$:

$$\mathrm{TC}(z_1, \ldots, z_n) \;=\; D_{\mathrm{KL}}\left(q(z_1, z_2, \ldots, z_n) \;\middle\|\; \prod_{j=1}^{n} q(z_j)\right). \tag{1.2}$$

When TC equals zero, all variables are mutually independent. Minimising TC during training therefore encourages a model to learn representations whose components carry non-overlapping information—the core idea behind *disentangled* representation learning (Section 1.3).

1.1.5 Mutual Information

The *mutual information* between two random variables X and Y quantifies how much knowing X reduces uncertainty about Y:

$$I(X;Y) \;=\; D_{\mathrm{KL}}\big(P_{XY} \| P_X \otimes P_Y\big) \;=\; \mathbb{E}_{(x,y)\sim P_{XY}}\left[\log \frac{P_{XY}(x,y)}{P_X(x)\,P_Y(y)}\right]. \tag{1.3}$$

In fair representation learning, we want the encoded features $\mathbf{z}$ to have *high* mutual information with the input $\mathbf{x}$ (to preserve utility) but *low* mutual information with a sensitive attribute s (to ensure fairness).

1.1.6 Evidence Lower Bound (ELBO)

Exact marginal likelihood $\log p_\Theta(\mathbf{x})$ is intractable for deep latent-variable models. Jensen's inequality gives a tractable surrogate known as the **Evidence Lower BOund (ELBO)**:

$$\log p_\Theta(\mathbf{x}) \;\geq\; \underbrace{\mathbb{E}_{q_\Phi(\mathbf{z}|\mathbf{x})}\big[\log p_\Theta(\mathbf{x}|\mathbf{z})\big]}_{\text{reconstruction term}} \;-\; \underbrace{D_{\mathrm{KL}}\big(q_\Phi(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\text{regularisation term}} \;=:\; \mathcal{L}_{\mathrm{VAE}}. \tag{1.4}$$

Maximising $\mathcal{L}_{\mathrm{VAE}}$ simultaneously encourages faithful reconstruction and a well-behaved latent distribution.

1.1.7 Conditional Value-at-Risk (CVaR)

Standard empirical risk minimisation (ERM) minimises the *average* loss:

$$\mathcal{R}_{\mathrm{avg}}(\theta) \;:=\; \mathbb{E}_{(X,Y)\sim P}\big[\ell(\theta;\, X, Y)\big]. \tag{1.5}$$

Because the average is dominated by easy (majority) samples, ERM tends to underperform on rare demographic subgroups—precisely the fairness problem. *Distributionally Robust Optimisation* (DRO) addresses this by minimising the *worst-case* risk $\mathcal{R}_{\max}(\theta) = \max_m \mathbb{E}_{(X,Y)\sim P_m}[\ell(\theta; X, Y)]$ over unknown latent groups.

A tractable upper bound on $\mathcal{R}_{\max}$ is given by *Conditional Value-at-Risk* at level $\alpha \in (0,1)$ [2]:

$$\mathrm{CVaR}_\alpha(\theta) \;:=\; \inf_{\lambda\in\mathbb{R}} \left\{ \lambda + \frac{1}{\alpha}\,\mathbb{E}_{(X,Y)\sim P}\big[\ell(\theta;\, X, Y) - \lambda\big]_+ \right\}, \tag{1.6}$$

where $[a]_+ = \max\{0, a\}$ is the hinge (ReLU) function.

Proposition (Proposition 1 in [12]). *If each latent group m occurs with probability $\pi_m \geq \alpha$, then* $\mathrm{CVaR}_\alpha(\theta) \geq \mathcal{R}_{\max}(\theta)$. Hence minimising CVaR minimises an upper

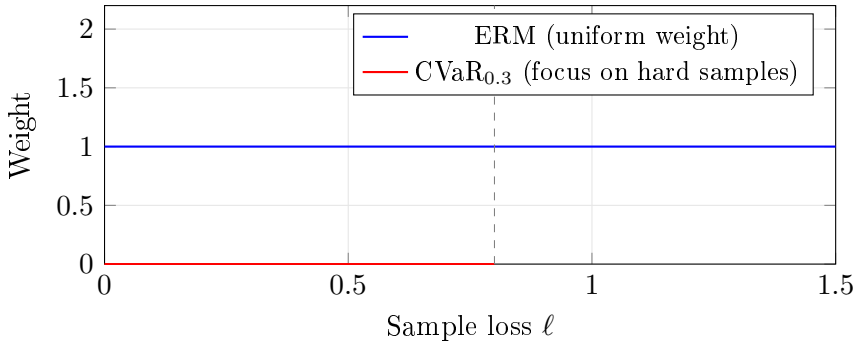

Figure 1.1: Comparison of sample weighting under ERM and CVaR at $\alpha = 0.3$. ERM weights all samples uniformly, whereas CVaR assigns zero weight to easy samples (loss $< \lambda^*$) and amplified weight to hard samples (loss $> \lambda^*$). Minimising CVaR therefore focuses training on the worst-performing subgroups.

bound on the worst-case group risk.

Group-level CVaR. When demographic labels are available, one can apply CVaR at the group level:

$$\mathrm{CVaR}_\alpha^{\mathcal{G}}(\theta) \;:=\; \inf_{\lambda \in \mathbb{R}} \left\{ \lambda + \frac{1}{\alpha} \mathbb{E}_{G \sim \mathrm{Uniform}(\mathcal{G})} \big[\mathcal{R}_G(\theta) - \lambda \big]_+ \right\}, \tag{1.7}$$

where $\mathcal{R}_G(\theta)$ is the per-group risk. This formulation simultaneously minimises average group risk and minimises the disparity across groups [13].

1.2 Representation Learning

Representation learning is the subfield of machine learning concerned with automatically discovering compact, informative features (representations) from raw data [3]. Rather than hand-crafting features, a model—typically a deep neural network—learns a function $f_\theta : \mathcal{X} \to \mathcal{Z}$ that maps high-dimensional inputs $\mathbf{x}$ to a lower-dimensional *latent* (embedding) vector $\mathbf{z}$. The key desiderata are:

1. **Informativeness**: $\mathbf{z}$ retains enough information about $\mathbf{x}$ for downstream tasks.
2. **Compactness**: $\mathbf{z}$ discards irrelevant variation (noise, style, sensitive attributes).
3. **Transferability**: representations learned on one task generalise to related tasks.

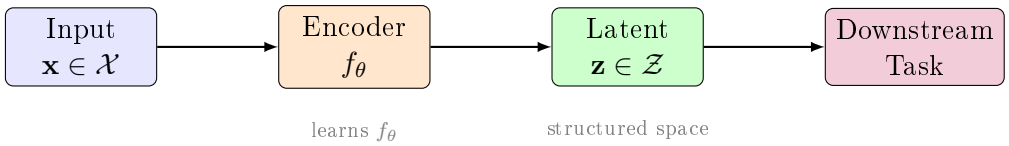

Figure 1.2: High-level picture of representation learning. The encoder f_θ maps a raw input $\mathbf{x}$ to a structured latent vector $\mathbf{z}$, which is then used for a downstream task such as classification or fairness-aware prediction.

1.3 Disentangled Representation Learning

Standard representation learning makes no assumption about the *structure* of the latent space. *Disentangled* representation learning imposes the additional constraint that different dimensions (or subspaces) of $\mathbf{z}$ capture statistically independent generative factors of the data [3–5].

Formally, if the data is generated by K independent factors $v_1, v_2, \ldots, v_K$, a disentangled encoder should produce a latent vector $\mathbf{z} = [z_1, z_2, \ldots, z_K]$ such that

$$q_\Phi(z_1, z_2, \ldots, z_K) \approx \prod_{k=1}^{K} q_\Phi(z_k),$$

i.e., the joint latent distribution factorises into independent marginals. This is precisely the condition that Total Correlation (Eq. 1.2) is zero.

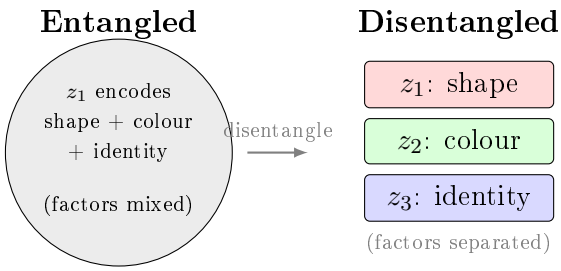

Figure 1.3: Conceptual illustration of disentangled vs. entangled latent representations. In an entangled space, each dimension conflates multiple generative factors. A disentangled space assigns separate, interpretable dimensions to each factor, enabling controlled manipulation and attribute-specific filtering.

Why does disentanglement matter for fairness? If a latent space is properly disentangled, one can isolate and *remove* the subspace that encodes sensitive demographic information before passing representations to a downstream classifier, achieving fairness without sacrificing task-relevant information.

1.4 Autoencoders

An *autoencoder* is a neural network trained to reconstruct its input through a bottleneck layer. It consists of two components:

- **Encoder** $E_\theta : \mathcal{X} \to \mathcal{Z}$ compresses the input $\mathbf{x}$ into a latent code $\mathbf{z} = E_\theta(\mathbf{x})$.
- **Decoder** $G_\phi : \mathcal{Z} \to \mathcal{X}$ reconstructs the input as $\hat{\mathbf{x}} = G_\phi(\mathbf{z})$.

Training minimises reconstruction loss, typically mean-squared error:

$$\mathcal{L}_{\text{AE}} \;=\; \mathbb{E}_{\mathbf{x}}\Big[\big\|\mathbf{x} - G_\phi(E_\theta(\mathbf{x}))\big\|^2\Big]. \tag{1.8}$$

The bottleneck forces the encoder to discard redundant information and retain only the structure necessary for reconstruction.

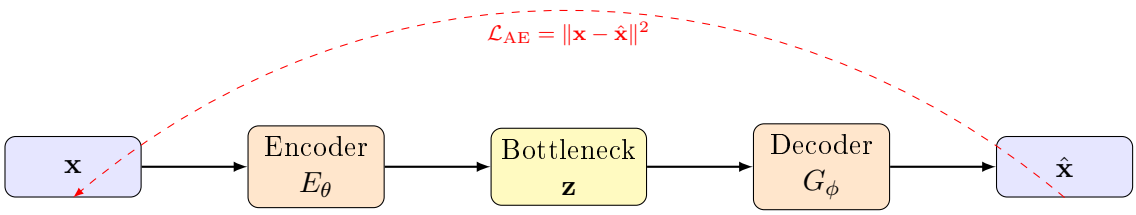

Figure 1.4: Architecture of a standard autoencoder. The encoder maps the input $\mathbf{x}$ to a compact latent code $\mathbf{z}$; the decoder reconstructs the input as $\hat{\mathbf{x}}$. The reconstruction error (red dashed arrow) drives training.

1.5 Variational Autoencoders (VAE)

A *Variational Autoencoder* (VAE) [7] is a probabilistic generative model that frames encoding and decoding as Bayesian inference. Instead of mapping an input to a single point in latent space, the encoder outputs a *distribution*:

$$q_\Phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}\big(\boldsymbol{\mu}_{q_\Phi}(\mathbf{x}),\ \boldsymbol{\sigma}^2_{q_\Phi}(\mathbf{x})\big),$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are output by the encoder network with parameters Φ. The decoder models $p_\Theta(\mathbf{x}|\mathbf{z})$.

Reparameterisation trick. To enable gradient-based optimisation through a stochastic node, we sample

$$\mathbf{z} = \boldsymbol{\mu}_{q_\Phi}(\mathbf{x}) + \boldsymbol{\sigma}_{q_\Phi}(\mathbf{x}) \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{1.9}$$

which moves the randomness outside the computational graph.

Training objective. The VAE is trained by maximising the ELBO (Eq. 1.4), which can be written explicitly as:

$$\mathcal{L}_{\text{VAE}} = \sum_{i=1}^{N} \Big[\mathbb{E}_{q_\Phi(\mathbf{z}|\mathbf{x}_i)} \big[\log p_\Theta(\mathbf{x}_i|\mathbf{z}) \big] - D_{\text{KL}} \big(q_\Phi(\mathbf{z}|\mathbf{x}_i) \,\|\, p(\mathbf{z}) \big) \Big]. \tag{1.10}$$

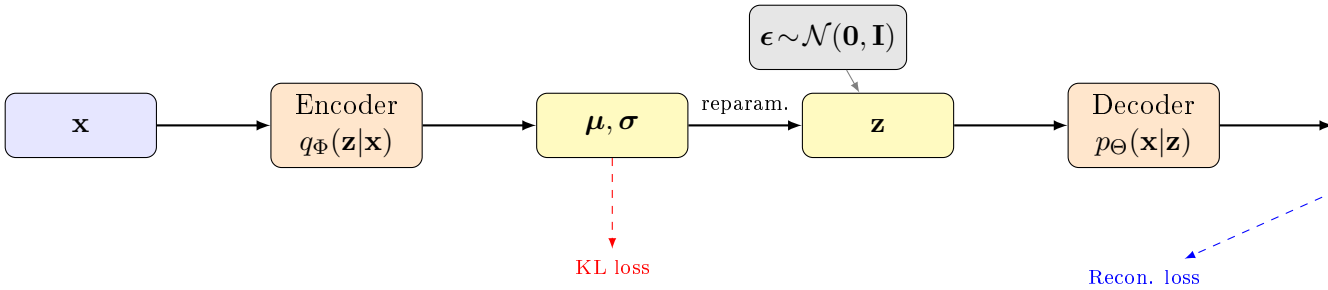

Figure 1.5: Variational Autoencoder (VAE) architecture. The encoder produces a Gaussian distribution $(\boldsymbol{\mu}, \boldsymbol{\sigma})$ in latent space; the reparameterisation trick yields a differentiable sample $\mathbf{z}$. The decoder reconstructs the input. Training balances a reconstruction loss (blue) and a KL-divergence regulariser (red).

Extensions for disentanglement. β-VAE [4] up-weights the KL term by a hyper-parameter $\beta > 1$ to encourage greater disentanglement at the cost of reconstruction quality. FactorVAE [5] and β-TCVAE [6] instead minimise Total Correlation directly (Eq. 1.2), yielding better disentanglement without degrading reconstruction.

1.6 Adversarial Learning

Adversarial learning is a training paradigm in which two model components compete in a minimax game, each driving the other towards a more useful solution [8].

1.6.1 Generative Adversarial Networks (GANs)

In the canonical GAN setup, a generator G learns to produce synthetic samples that fool a discriminator D, while D learns to distinguish real from fake:

$$\min_G \max_D \; \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \big[\log D(\mathbf{x}) \big] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} \big[\log(1 - D(G(\mathbf{z}))) \big]. \tag{1.11}$$

At equilibrium, G produces samples from p_{data} and D cannot distinguish real from generated samples.

1.6.2 Adversarial Debiasing

In fair representation learning, adversarial networks serve a different role. A *predictor* P is trained to predict a target label y from a representation $\mathbf{z}$, while an *adversary* A attempts to predict the sensitive attribute s from the same $\mathbf{z}$. The encoder is then trained to *fool the adversary*:

$$\min_{E,P} \max_{A} \mathcal{L}_{\text{pred}}(P(E(\mathbf{x})),\, y) \;-\; \lambda\, \mathcal{L}_{\text{adv}}(A(E(\mathbf{x})),\, s). \tag{1.12}$$

The negative sign on the adversarial loss means the encoder *maximises* the adversary's error—equivalently, it removes sensitive information from $\mathbf{z}$.

Gradient Reversal Layer (GRL). A practical implementation of Eq. 1.12 uses a Gradient Reversal Layer (GRL) [9]. During the forward pass, GRL acts as the identity; during back-propagation, it negates the gradient, automatically implementing the minimax objective without alternating optimisation:

$$\text{GRL}(\mathbf{z}) = \mathbf{z}, \qquad \frac{\partial\, \text{GRL}(\mathbf{z})}{\partial\, \mathbf{z}} = -\lambda_{\text{grl}}\, \mathbf{I}. \tag{1.13}$$

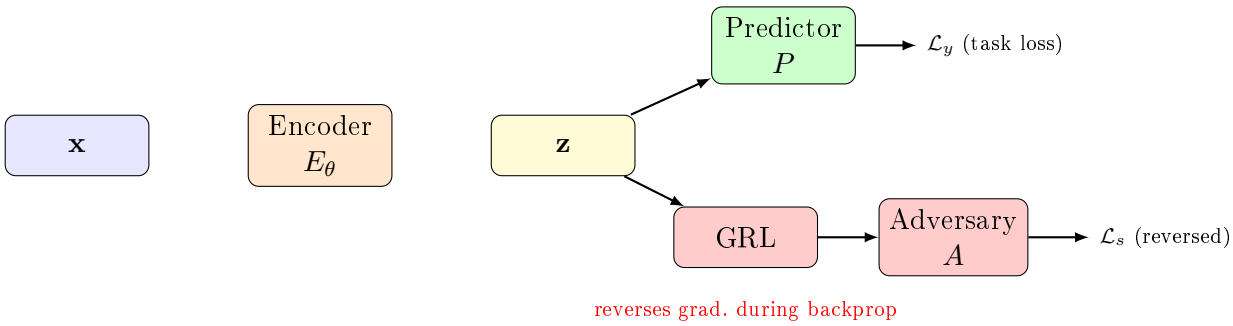

Figure 1.6: Adversarial debiasing with a Gradient Reversal Layer (GRL). The predictor branch optimises task performance on target label y; the adversary branch tries to recover sensitive attribute s from $\mathbf{z}$. The GRL reverses the adversary gradient, training the encoder to suppress demographic information in $\mathbf{z}$.

1.7 Fractional Fourier Transform (FrFT)

The classical Fourier transform maps a signal from the time domain to the frequency domain. The *Fractional Fourier Transform* (FrFT) [14] is a generalisation parameterised by a rotation order $\alpha \in [0, 1]$, interpolating continuously between the time ($\alpha = 0$) and frequency ($\alpha = 1$) domains:

$$\mathcal{F}_\alpha[f](u) = \int_{-\infty}^{\infty} f(t)\, K_\alpha(t, u)\, \mathrm{d}t, \tag{1.14}$$

where the kernel is

$$K_\alpha(t,u) = A_\alpha \exp\bigl(i\pi(t^2\cot\alpha - 2tu\csc\alpha + u^2\cot\alpha)\bigr), \qquad A_\alpha = \sqrt{1 - i\cot\alpha}.$$

Deepfake generation pipelines introduce subtle frequency artefacts that are more discriminative in intermediate spatial–frequency domains than in either the pure spatial or pure frequency domain [15]. By selecting α between 0 and 1, the FrFT exposes these artefacts while reducing reliance on demographic appearance features (skin tone, facial structure) that dominate the spatial domain.

1.8 Fairness Concepts in Machine Learning

Before defining evaluation metrics, we establish the terminology used throughout this report.

- **Target attribute** (y_t): the label the model is trained to predict (e.g., "attractive", "real/fake").
- **Protected (sensitive) attribute** (y_p): a demographic characteristic (e.g., gender, race, age) with respect to which the model should *not* discriminate.
- **Group fairness**: equal treatment of pre-specified demographic groups.
- **Individual fairness**: similar individuals receive similar predictions.

Let $\hat{Y}$ denote the model's prediction, Y the ground truth label ($Y = 1$ positive, $Y = 0$ negative), and $G \in \{g_0, g_1\}$ the protected group membership. Define:

$$\begin{aligned}
\mathrm{TPR}_g &= \Pr(\hat{Y} = 1 \mid Y = 1,\, G = g), && \text{(True Positive Rate for group } g\text{)}\\
\mathrm{TNR}_g &= \Pr(\hat{Y} = 0 \mid Y = 0,\, G = g), && \text{(True Negative Rate for group } g\text{)}\\
\mathrm{FPR}_g &= \Pr(\hat{Y} = 1 \mid Y = 0,\, G = g). && \text{(False Positive Rate for group } g\text{)}
\end{aligned}$$

1.9 Evaluation Metrics

1.9.1 Equal Opportunity

Equal Opportunity [10] requires that the True Positive Rate be equal across protected groups. A model satisfies equal opportunity if subjects who *deserve* a positive outcome are equally likely to receive it, regardless of group membership. It is measured as:

$$\Delta_{\mathrm{EO}} = \bigl|\mathrm{TPR}_{g_0} - \mathrm{TPR}_{g_1}\bigr|. \tag{1.15}$$

A value of $\Delta_{\mathrm{EO}} = 0$ indicates perfect equal opportunity; *lower is better.*

1.9.2 Equalized Odds

Equalized Odds [10] is a stricter criterion: the model must achieve equal TPR *and* equal TNR across groups simultaneously. This ensures both that qualified candidates are equally recognised and that unqualified ones are equally rejected:

$$\Delta_{\text{EOdd}} = \frac{1}{2}\Big(\big|\text{TPR}_{g_0} - \text{TPR}_{g_1}\big| + \big|\text{TNR}_{g_0} - \text{TNR}_{g_1}\big|\Big). \tag{1.16}$$

Again, *lower is better*. The relationship between these two metrics is: a model satisfying equalized odds also satisfies equal opportunity, but not vice versa.

1.9.3 Standard Accuracy

Standard accuracy is the fraction of correctly classified samples:

$$\text{Acc} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}. \tag{1.17}$$

While essential for gauging utility, accuracy alone is a misleading fairness proxy. On an imbalanced dataset biased towards one demographic group, a model can achieve high accuracy while systematically harming an underrepresented group.

1.9.4 Equalized Accuracy

To address the limitations of standard accuracy on skewed test distributions, the README paper [11] introduces *Equalized Accuracy*, which averages TPR and TNR across both groups equally:

$$\text{EAcc} = \frac{1}{4}\Big(\text{TPR}_{g_0} + \text{TNR}_{g_0} + \text{TPR}_{g_1} + \text{TNR}_{g_1}\Big). \tag{1.18}$$

Because each group contributes equally regardless of its size in the dataset, Equalized Accuracy is a more reliable indicator of a model's generalisation across the population.

1.9.5 FPR-Based Fairness Metrics for Deepfake Detection

In the context of deepfake detection, where real faces vastly outnumber deepfakes in practice, *False Positive Rate* disparities can have severe societal consequences: real individuals being misclassified as deepfakes and subjected to unjust scrutiny [12]. Three metrics capture this:

1. **FPR Gap** (G_{FPR}): maximum pairwise difference in FPR across demographic

groups:

$$G_{\text{FPR}} = \max_{g,g' \in \mathcal{G}} \left| \text{FPR}_g - \text{FPR}_{g'} \right|. \tag{1.19}$$

2. **Equal FPR** (F_{FPR}): summed absolute difference between each group's FPR and the overall FPR:

$$F_{\text{FPR}} = \sum_{g \in \mathcal{G}} \left| \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1,\, G_i = g,\, Y_i = 0]}{\sum_i \mathbf{1}[G_i = g,\, Y_i = 0]} - \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1,\, Y_i = 0]}{\sum_i \mathbf{1}[Y_i = 0]} \right|. \tag{1.20}$$

3. **Fair Equalized Odds** (F_{EO}): extends Equal FPR to both positive and negative classes:

$$F_{\text{EO}} = \sum_{g \in \mathcal{G}} \sum_{q \in \{0,1\}} \left| \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1,\, G_i = g,\, Y_i = q]}{\sum_i \mathbf{1}[G_i = g,\, Y_i = q]} - \frac{\sum_i \mathbf{1}[\hat{Y}_i = 1,\, Y_i = q]}{\sum_i \mathbf{1}[Y_i = q]} \right|. \tag{1.21}$$

For all three metrics, lower values indicate greater fairness.

1.9.6 Summary of Metrics

Table 1.1 summarises all evaluation metrics, their target conditions, and the direction of improvement.

Table 1.1: Summary of fairness and utility evaluation metrics used in this report. All fairness metrics are minimised; accuracy metrics are maximised.

Metric	**Symbol**	**Condition checked**	**Better**
Equal Opportunity	Δ_{EO}	TPR equal across groups	$\downarrow$
Equalized Odds	Δ_{EOdd}	TPR & TNR equal	$\downarrow$
Standard Accuracy	Acc	Overall correctness	$\uparrow$
Equalized Accuracy	EAcc	Balanced group accuracy	$\uparrow$
FPR Gap	G_{FPR}	Max FPR spread	$\downarrow$
Equal FPR	F_{FPR}	FPR parity per group	$\downarrow$
Fair Equalized Odds	F_{EO}	TPR & FPR parity	$\downarrow$
AUC-ROC	AUC	Ranking quality	$\uparrow$

1.10 Key Deep Learning Architectures Used in Fair AI

Table 1.2 provides a concise comparison of the core neural-network models referenced throughout this report.

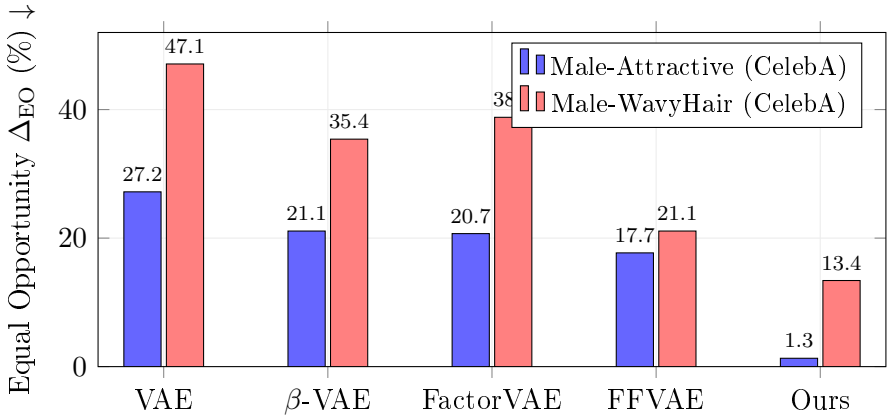

Figure 1.7: Equal Opportunity scores (Δ_{EO}, lower is better) on the CelebA dataset for two attribute pairs. Data from the README paper [11]. The proposed FD-VAE ("Ours") substantially outperforms all baselines, demonstrating that explicit disentanglement of the protected-attribute subspace dramatically reduces group-wise TPR disparities.

1.11 Chapter Summary

This chapter introduced the mathematical and conceptual vocabulary needed to follow the technical discussions in subsequent chapters.

- **Mathematical tools**: KL divergence, Total Correlation, mutual information, ELBO, and CVaR provide the optimisation-theoretic backbone for fair representation learning.
- **Architectures**: Autoencoders, VAEs, and adversarial networks are the primary building blocks; disentanglement methods extend them to produce interpretable, attribute-separable latent spaces.
- **FrFT**: Offers a hybrid spatial–frequency domain well-suited to capturing deepfake artefacts that are less correlated with demographic features.
- **Evaluation metrics**: Equal Opportunity, Equalized Odds, Equalized Accuracy, and FPR-based metrics together give a multi-dimensional picture of fairness; no single metric suffices.

With these tools in hand, Chapter 3 examines the README (FD-VAE) architecture, which was the first method to explicitly address the three-way attribute entanglement problem in fair representation learning.

Table 1.2: Key model architectures referenced in the fairness-in-AI literature covered in this report.

Model	**Core idea**	**Strength**	**Limitation**
VAE [7]	Probabilistic encoder–decoder with Gaussian latent prior	Smooth, generative latent space	No explicit control over which factors are captured
β-VAE [4]	Up-weight KL term by β	Encourages disentanglement	Hurts reconstruction at high β
FactorVAE [5]	Minimise Total Correlation via discriminator	Better disentanglement than β-VAE	Requires an additional discriminator
FFVAE [16]	Disentangle sensitive vs. non-sensitive latents	Fairness-focused VAE	Sensitive latents leak target attribute info
FD-VAE [11]	Three-way split: TAL, PAL, MAL + decorrelation loss	Explicitly decorrelates attributes	Requires target & protected attribute labels
Adversarial Autoencoder	GRL to strip demographic info from $\mathbf{z}$	Architecture-level fairness	Adversarial training instability
FAIR-FRFT (ours)	FrFT + adversarial autoencoder + CVaR	Frequency-domain fairness	Higher computational cost

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